
Predictive Evaluation Competition

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Abstract

We address the Predictive AI Evaluation Challenge, a cold-start problem in which a model must predict $\hat{p} = P(\text{correct} \mid \text{subject, item})$ given only four text fields and no prior observations of the test items. Our pipeline combines a 2PL Item Response Theory backbone with a sentence-transformer encoder (all-mpnet-base-v2)—shared across subject, item, benchmark, and condition—feeding a 5-layer MLP head; logits then pass through post-hoc Platt and temperature scaling, and the $K=5$ adaptive labels are folded in as a single per-subject mean-residual offset. Our first leaderboard submission *regressed* to -1.01 NLL, a 0.31 NLL drop from the team’s earlier submission at commit 40976e8 (-0.70); three diagnostic submissions and a single scalar temperature ($T \approx 4.07$, fit on a leave-two-benchmarks-out hold-out) recover the curve to -0.65 , and an equal-weight arithmetic blend with the per-subject training-time mean accuracy ($0.5 p_{T\text{-scaled}} + 0.5 \bar{y}_s$) reaches -0.61 ; AUC rises in lockstep from 0.63 (content-only) to 0.69 ($\alpha=0.3$ blend), so per-subject identity buys both calibration *and* ranking. A Part V coverage sweep reveals a sharper version of the same story: a 745-key *lookup* of the smoothed subject mean (no encoder, no MLP, no calibration) also scores -0.61 with AUC 0.68 , tying the full blend; the content head contributes *ranking* but no detectable NLL once per-subject identity is available. We argue that for cold-start $P(\text{correct})$ prediction the dominant lever is per-subject prior + controlled under-confidence at test time, not encoder choice.¹

1 Problem Formulation & Method Description

1.1 Problem Setting

Let $\mathcal{S}$ be subjects (language-model checkpoints) and $\mathcal{I}$ items (benchmark questions). Training data from aims-foundations/measurement-db ? forms a sparse response matrix $R \in \{0, 1, \text{NaN}\}^{|\mathcal{S}| \times |\mathcal{I}|}$ where $R_{ij} = 1$ iff subject i answers item j correctly under condition c . At test time the scorer asks for $\hat{p} \in (0, 1)$ on items whose benchmarks do *not* appear in training, scored by mean Bernoulli log-likelihood (higher is better; $\hat{p}=0.5$ gives $-\log 2 \approx -0.693$). The platform optionally reveals $K=5$ ground-truth labels per round (chosen by our `acquisition_function`) *before* the prediction round on the remainder—the only direct observation of the held-out distribution available at inference.

1.2 Method Overview

The pipeline factorizes into a low-rank *ability backbone* and a content-aware *residual head*, in the spirit of factor-augmented psychometric regression. (i) A 2PL IRT model fit on the dense submatrix (≥ 5 observed responses; $\sim 33\text{K}$ items) yields a subject ability θ_i and per-item (a_j, b_j) . (ii) A shared all-mpnet-base-v2 ?? encoder embeds all four text fields into 768-d vectors, so that benchmarks and conditions unseen at training time still receive meaningful representations rather than the all-zero

¹Code: <https://github.com/daphne-barretto/prediction-evaluation-challenge>.

categorical indicator a one-hot scheme would produce. (iii) A 5-layer MLP head maps the 3075-d concatenation $[\theta, \mathbf{e}_{\text{subj}}, \mathbf{e}_{\text{item}}, \mathbf{e}_{\text{bm}}, \mathbf{e}_{\text{cond}}, \mu_{\text{subj}}, \mu_{\text{bm}}]$ to a logit (architecture in Appendix B.1; $\mu_{\text{subj}}, \mu_{\text{bm}}$ are empirical pass-rates). (iv) Post-hoc Platt re-scaling $\sigma(\alpha\ell + \beta)$? fit on a 10% in-distribution hold-out, then temperature scaling $\sigma(\ell/T)$? fit on a cold-start hold-out. (v) At inference the $K=5$ labels are folded in as a single per-subject mean-residual offset $\delta_s = \frac{1}{K} \sum_k (\hat{p}_k - y_k)$, subtracted from subsequent predictions; outputs are clipped to $[10^{-3}, 1-10^{-3}]$.

Adaptive labeling. We pick the K items that maximize $\text{score}(x) = (1 - |2\hat{p}(x) - 1|) \cdot \frac{1}{1+n_{c(x)}}$, the product of model uncertainty and a diversity penalty over 64 pre-fit k -means centroids on item embeddings. Either factor alone is degenerate (uncertainty collapses semantically; diversity wastes labels on easy items); the product favors items that are both informative and representative of distinct regions of the input distribution. Revealed labels feed the offset above. We leave θ unchanged at test time because $K=5$ Newton–Raphson updates were empirically unstable (Appendix A.2).

Training procedure. We fit on $\sim 2.7\text{M}$ response rows held out from `bfc1` and `matharena`; total wall-clock time is $\approx 20\text{--}25$ min on a single T4 GPU. Full pipeline (compute, data binarization, artifact list) in Appendix B.2.

2 Training Dataset

We use the public `aims-foundations/measurement-db` parquet shards ? (no additional curated data). After join and binarization, the usable training set contains $\sim 3.4\text{M}$ response rows over 909 subjects, $\sim 104\text{K}$ items, 15 benchmarks (13 train, 2 cold-start hold-out: `bfc1` and `matharena`), and 48 conditions; overall pass-rate ~ 0.65 . Of those items, $\sim 33\text{K}$ have ≥ 5 recorded responses and participate in the dense IRT fit; the remainder receive the mean (a, b) as fallback. Downstream-split row counts are in Table 6 (Appendix B.3); detailed preprocessing, label binarization, and cold-start split construction are also in Appendix B.3.

3 Experimental Results & Ablation Studies

3.1 Validation Setup

We use two complementary validation regimes plus the leaderboard. The *in-distribution Platt hold-out* is a 10% uniform random hold-out of non-cold-start rows on which Platt (α, β) are fit; we also report its post-Platt NLL as an in-distribution calibration sanity check. The *cold-start hold-out* removes a fixed pair of benchmarks (`bfc1`, `matharena`) entirely from training to estimate the OOD gap, since their items, conditions, and item families are unseen at training time. We additionally report leaderboard NLL on Codabench.

3.2 Main Results

Table 1 reports NLL across the three regimes plus four diagnostic Codabench submissions that progressively disable test-time components and then add post-hoc temperature scaling ($T=4.073$ fit on the cold-start hold-out).

Headline finding. Sub 1 fell to -1.01 , *worse* than constant $\hat{p}=0.5$. Removing Platt (sub 2) and the per-subject offset (sub 3) recover 0.16 NLL; post-hoc T-scaling (sub 5) adds a further 0.20, landing at -0.65 — 0.05 NLL above the team’s earlier 40976e8 submission (-0.70) and 0.07 NLL below the public top entry (-0.58). AUC is stable at 0.62–0.63 across all four submissions — the entire 0.36-NLL improvement comes from *calibration*, not from any change to the ranking signal. The temperature was fit on the cold-start hold-out (-0.665) and generalised almost perfectly to the leaderboard (-0.650); a 0.015 NLL gap.

Equivalence of three calibration families. We compared three one- and two-parameter calibrators on the same cold-start hold-out ($N \approx 4 \times 10^5$ logits): pure T-scaling, prior shrinkage $\alpha\sigma(\ell) + (1-\alpha)/2$, and their composition. All three converge to within ~ 0.001 NLL ($T^*=4.07 \Rightarrow -0.6648$; $\alpha^*=0.386 \Rightarrow -0.6643$; joint -0.6639). Full T -vs- α contour in Appendix A.1.

Table 1: NLL and ROC AUC by evaluation regime and ablation submission. “offset” is the $K=5$ per-subject mean-residual; “T-scaling” applies $\sigma(\text{logit}/T)$ with $T=4.073$. Reference is $-\log 2 \approx -0.693$.

Regime / Submission	NLL	AUC
Constant prediction $\hat{p} = 0.5$	-0.693	0.500
In-distribution Platt hold-out (Platt)	-0.370	—
Cold-start hold-out, {bfcl, matharena} (pre-Platt)	-0.832	—
Cold-start hold-out, {bfcl, matharena} (Platt)	-0.884	—
Cold-start hold-out, {bfcl, matharena} (T-scaling, $T=4.073$)	-0.665	—
Codabench: Platt + offset (sub 1, round 742341)	-1.012	0.620
Codabench: identity-Platt + offset (sub 2)	-0.930	0.630
Codabench: identity-Platt, no offset (sub 3)	-0.850	0.630
Codabench: 40976e8 reproduction (sub 4)	-0.820	0.580
Codabench: identity-Platt, no offset, T-scaling (sub 5)	-0.650	0.630
Earlier team submission (40976e8, MiniLM-L3 + IRT)	-0.700	—
Top of public leaderboard (reference)	-0.580	—

3.3 Ablation Studies

We also have the team’s earlier 40976e8 submission (izhan branch) which shipped a MiniLM-L3 encoder, one-hot benchmark/condition features, and a Newton–Raphson θ update; it scored -0.70 on the leaderboard. The component-by-component delta (Table 3, Appendix A.2) shows that the encoder, feature scheme, MLP shape, calibration, and K -shot use all changed simultaneously. The submission climb of §3.2 then attributes the recovery to the calibration column of that table: removing Platt, removing the per-subject offset, and finally adding T-scaling are sufficient to reach -0.65 , 0.05 NLL above the 40976e8 submission, without touching the encoder upgrade or the text-embedded categorical scheme.

3.4 Strategy coverage: the calibration climb and family map

To bound how much of the leaderboard NLL is recoverable from calibration alone, we ran a deliberate set of ablations spanning the two scaffolds in the Discussion 6 tutorial: a seven-rung calibration climb (R1: $\hat{p}=0.5$ to R7: full PGE with $K=5$ adaptive labels) and a six-family option space (**F1** priors, **F2** IRT alone, **F3** content, **F4** metadata, **F5** local LLM judge, **F6** adaptive). Table 2 maps each submission onto these axes; per-submission leaderboard scores are in Table 4 (Appendix A.3).

Table 2: Coverage of Discussion 6 strategy axes. Rows are the seven-rung calibration climb; columns are the six families. ✓ marks ablations we shipped; × marks families we deliberately did not pursue (F2 IRT-only fails cold-start by design ?; F5 local LLM judge does not fit the per-tier sandbox compute budget).

	F1 priors	F2 IRT-only	F3 content	F4 metadata	F5 local LLM	F6 adaptive
R1: $\hat{p} = 0.5$	✓					
R2: global mean	✓					
R3: smoothed subject mean	✓			✓		
R4: per-(bench, cond) mean	✓			✓		
R5: + content (text $\rightarrow$ params)			✓		×	
R6: + IRT θ (full PGE)		×	✓			
R7: + adaptive $K=5$ labels			✓			✓

Findings (full results in Appendix A.3). The R1–R4 floors expose a clean NLL decomposition: sub 9 ($\hat{p}=0.5$) scores -0.69 ; sub 10 ($\hat{p}=0.645$, global mean) scores -0.65 , *tying the T-scaled MLP pipeline*; sub 21 (subject-mean lookup, no encoder/MLP) scores -0.61 with AUC 0.68; sub 22

(per-benchmark mean) hits the constant-fallback floor at NLL -0.65 , AUC 0.50. The 0.08 NLL budget over $-\ln 2$ splits into 0.04 from the population mean and 0.04 from per-subject identity. **Sub 17** ($\alpha=0.3$ blend) wins at $-0.61/\text{AUC } 0.69$: a concave AUC peak on a flat NLL plateau across $\alpha \in [0, 0.5]$ (subs 17/21/13/25); the content head buys ranking but no NLL. Sub 20 (sub 13 + K-shift) scores -0.63 , confirming the $K=5$ residual is too noisy.

3.5 Leaderboard Performance

We submitted 26 models. Sub 1 (mpnet_platt_meanresid) returned -1.012 (dissected in §4.1); subs 2, 3, 5 recover to -0.65 ; **sub 17** reaches $-0.61/0.69$ (subs 13, 21, 23, 24, 25, 26 tie the NLL floor). The remaining variants extend the Part V family map (Appendix A.3).

4 Analysis of Failure Modes & Interesting Patterns

4.1 Error Analysis

Ranking versus calibration. An AUC of 0.62 on the Codabench test set means sub 1 still orders pairs of (correct, incorrect) items correctly 62% of the time. The -1.01 NLL is therefore a *confidence* failure, not a discrimination failure: confidently-wrong predictions cost heavily under log-loss but little under AUC. We attribute the collapse to three compounding mechanisms (detailed in Appendix A.4): Platt sharpening $\alpha \approx 1.15$ pushes confidently-wrong OOD predictions toward $\{0, 1\}$; the encoder upgrade trades implicit-regularisation capacity for tighter in-distribution fit; and the per-subject mean-residual offset is exquisitely sensitive to which 5 items the platform happens to reveal.

4.2 Interesting Patterns

Distribution shift dominates the error budget. Going from a 10% random hold-out (NLL -0.37) to a leave-two-benchmarks-out hold-out (-0.88) to the Codabench test set (-1.01 for sub 1) shows a clean monotonic relationship between distribution shift and NLL — a gap of ≈ 0.6 NLL that cross-validation on the training distribution would have completely missed.

Calibration, not representation, is the binding lever. The single change that recovers the leaderboard score is post-hoc temperature scaling: parameter-free at training time, costs ~ 10 s to fit on the cold-start hold-out, and lands at -0.65 (sub 5). The encoder upgrade and the text-embedded categoricals are kept across all recovered submissions, yet contribute essentially nothing to the climb $3 \rightarrow 5$ — the entire 0.20 NLL recovery comes from a single scalar $T \approx 4$. The empirical equivalence of T-scaling, prior shrinkage, and their composition (§3.2) extends this: *any* 1- or 2-parameter post-hoc calibrator wins as long as it shrinks predictions toward the prior on the cold-start hold-out.

4.3 Limitations

- *Cold-start gap is large.* Even with text-embedded benchmarks, our pipeline degrades from -0.37 (in-distribution) to -0.83 (cold-start) to -1.01 (Codabench) before calibration — the MLP has not learned a truly transferable map from item text to correctness.
- *No encoder fine-tuning.* We use mpnet zero-shot; fine-tuning on $(item, label)$ pairs is the natural next step but exceeded the compute budget.
- *$K=5$ is a hard signal-to-noise floor.* A single global scalar offset (sub 14) is the most we trust ourselves to extract from 5 labels per category; any per-subject or per-cluster split is empirically unstable (sub 8 regressed by 0.10 NLL).

5 Conclusion

We treated the competition as a calibration problem dressed up as representation learning. Sub 17 ($\alpha=0.3$ blend) reached -0.61 NLL / AUC 0.69; sub 21 (745-key subject-mean lookup) hit $-0.61/0.68$ with no encoder, MLP, or calibration. The 0.08 NLL over $-\ln 2$ splits into 0.04 from the global mean and 0.04 from per-subject identity; the content head buys AUC but no NLL, and $K=5$ adds nothing under a Bayesian posterior. The 0.03 to the public top must come from item-text signal.

A Additional Results

A.1 Full T -vs- α calibration map

We dumped $N \approx 4 \times 10^5$ raw MLP logits from the trained sub 5 pipeline on the leave-two-benchmarks-out hold-out (bfc1+matharena) and swept three one- and two-parameter post-hoc calibrators by exact NLL minimization:

- **Temperature scaling** $\hat{p} = \sigma(\ell/T)$, $T \in [0.5, 10]$. The optimum is $T^* = 4.073$ with $\text{NLL} = -0.6648$.
- **Prior shrinkage** $\hat{p} = \alpha \sigma(\ell) + (1 - \alpha)/2$, $\alpha \in [0, 1]$. The optimum is $\alpha^* = 0.386$ with $\text{NLL} = -0.6643$.
- **Composition** $\hat{p} = \alpha \sigma(\ell/T) + (1 - \alpha)/2$ jointly in (T, α) . The optimum is $(T, \alpha) \approx (2.1, 0.6)$ with $\text{NLL} = -0.6639$.

All three converge to within 0.001 NLL of each other, and the (T, α) loss surface is essentially a single ridge: any pair on the line $T \cdot (2\alpha/(1 + \alpha)) \approx 1.6$ gives near-optimal NLL. The leaderboard sweep (subs 5–7, $T \in \{3.0, 4.07, 5.5\}$, all scoring -0.65) sits squarely on this ridge.

A.2 Component ablation: 40976e8 submission vs. sub 5

Table 3 compares the team’s earlier 40976e8 submission (izhan branch, leaderboard -0.70) against the sub 5 pipeline. Five components changed simultaneously; the diagnostic submission climb of §3.2 attributes the $+0.05$ NLL recovery to the calibration column alone.

Table 3: Component-by-component delta: team’s earlier 40976e8 submission vs. sub 5.

Component	40976e8 sub	Sub 5
Text encoder	paraphrase-MiniLM-L3-v2 (17M, 384-d)	all-mpnet-base-v2 (110M, 768-d)
Benchmark/condition	one-hot	text-embedded (shared encoder)
MLP head	4-layer 256-128-64-1	5-layer 512-256-128-64-1, LN+dropout
Calibration	none	Platt (identity in sub 5) + T-scaling ($T=4.07$)
$K=5$ use	Newton–Raphson on θ	per-subject mean residual (disabled in sub 5)
Leaderboard NLL	-0.70	-0.65

Why each change was expected to help.

- *Encoder upgrade.* all-mpnet-base-v2 is a more accurate sentence embedder on benchmark NLI tasks; we expected it to give better semantic separation between content categories.
- *Text-embedded categoricals.* One-hot indicators are all-zero for benchmarks unseen at training time; text embeddings give a non-degenerate representation that should at least transfer zero-shot.
- *Wider MLP.* 110M-d encoder outputs deserve more head capacity than a 4-layer 256-width path; LayerNorm + dropout were added to control the corresponding overfitting risk.
- *Platt re-calibration.* A logistic re-scaler on raw logits is the textbook fix for over- or under-confident classifiers.
- *Mean-residual offset.* Replaces the prior Newton–Raphson θ update with something that does not assume the 5-shot labels are unbiased samples from the subject’s overall accuracy.
- *T-scaling (post-hoc, sub 5).* A single scalar ($T=4.07$) fit on a leave-two-benchmarks-out hold-out; this is the change that actually moved the leaderboard.

A.3 Full Part V leaderboard sweep

Table 4 reports leaderboard NLL for each Part-V variant. The seven-rung calibration climb (R1–R7) is read from the slide-29 scaffold; subs 6, 7, 8 are robustness sweeps over T and the offset; subs 13–16 are the family-coverage variants.

Table 4: Part V ablation grid. “LB NLL” is the Codabench leaderboard score; “AUC” is the Codabench leaderboard AUC where reported; “-” indicates not (yet) returned by the platform.

Sub	Variant	Rung / Family	LB NLL	AUC
9	$\hat{p} \equiv 0.5$	R1 (F1 priors)	-0.69	0.50
10	global pass-rate 0.645	R2 (F1 priors)	-0.65	0.50
21	smoothed subject-mean lookup	R3 (F1+F4)	-0.61	0.68
22	per-benchmark mean lookup	R4 (F1+F4)	-0.65	0.50
5	T-scaled mpnet, $T=4.07$	R6/R5 (F3 content)	-0.65	0.63
6	T-scaled mpnet, $T=3.0$	R6 robustness	-0.65	0.63
7	T-scaled mpnet, $T=5.5$	R6 robustness	-0.65	0.63
8	sub 5 + per-subject offset	R7 (F3+F6)	-0.75	0.63
13	$0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$	R7 (F3+F4 blend)	-0.61	0.67
14	sub 5 + shrunk global mean-residual	R7 (F3+F6 light)	-0.65	0.63
17	$0.3 p_{\text{sub } 5} + 0.7 \bar{y}_s$	blend α sweep	-0.61	0.69
18	$0.7 p_{\text{sub } 5} + 0.3 \bar{y}_s$	blend α sweep	-0.62	0.66
19	$0.5 p_{\text{sub } 5} + 0.5 \mu_{\text{global}}$	prior-pull control	-0.63	0.63
20	sub 13 + shrunk global mean-residual	blend + K=5 shift	-0.63	0.67
25	$0.4 p_{\text{sub } 5} + 0.6 \bar{y}_s$	blend α sweep	-0.61	0.68
23	sub 21 with $\lambda=0.05$ shrinkage	sub 21 sweep	-0.61	0.68
24	sub 21 with $\lambda=0.20$ shrinkage	sub 21 sweep	-0.61	0.68
26	sub 21 + Beta-Bernoulli K=5 posterior	sub 21 + K likelihood	-0.61	0.68

R1–R4 (subs 9, 10, 21, 22). Sub 9 ($\hat{p} \equiv 0.5$) scored -0.69 , matching $-\ln 2 \approx -0.693$ to within rounding; sub 10 ($\hat{p} \equiv 0.645$, the global train mean) scored **-0.65** , *tying the entire T-scaled MLP pipeline*; **sub 21 (smoothed subject-mean lookup) scored -0.61 , tying the best blend (sub 13) using a 745-key training-time lookup table with no encoder, no MLP, and no calibration**; sub 22 (per-benchmark mean) scored -0.65 , identical to the sub 10 global-mean floor because every cold-start benchmark falls through to the global fallback. The full NLL-improvement budget over the $-\ln 2$ floor thus decomposes cleanly into 0.04 NLL from knowing the population mean (R1→R2) and a further 0.04 NLL from knowing per-subject identity (R2→R3); the content head and calibrator contribute nothing detectable on the leaderboard once per-subject identity is available.

T-robustness (subs 5, 6, 7). All three scored -0.65 on the leaderboard. This confirms the broad-ridge result from Appendix A.1: the leaderboard distribution is within the calibration manifold of our cold-start hold-out, and sub-percent-NLL gains are not achievable by fine-tuning T .

Sub 8 post-mortem. Re-enabling the per-subject mean-residual offset on top of T-scaling lost 0.10 NLL. With only $K=5$ labels per category the offset is too noisy: a single subject whose 5 revealed items happen to be hard pushes *all* of that subject’s predictions far from 0.5, where they incur disproportionate log-loss. Sub 14 was designed as a conservative response: aggregate all $K \times$ categories labels into a single global shift, shrink it by $\lambda=25$ ridge, and cap magnitude at $\pm 5\text{pp}$. Sub 14 scored -0.65 on the leaderboard, a no-op on top of sub 5: with T-scaled predictions already sitting near 0.5, the mean residual is dominated by noise.

Sub 13. The blend $0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$ scored -0.61 , our new best by 0.04 NLL. The mechanism is variance reduction: T-scaled mpnet outputs sit near 0.5 (high entropy, low ranking signal) and the per-subject training-time mean $\bar{y}_s$ is a zero-variance estimate of overall accuracy on the items the subject has already answered. Equal-weight arithmetic averaging preserves most of sub 5’s ranking while pulling each prediction toward a per-subject prior, which is exactly the configuration that under-confident, well-ranked predictors benefit from.

Sub 13 ablation sweep (subs 17–20, 25). To attribute sub 13’s gain we built five follow-ups: subs 17 ($\alpha=0.3$), sub 25 ($\alpha=0.4$), and sub 18 ($\alpha=0.7$) sweep the blend weight around sub 13’s $\alpha=0.5$; sub 19 replaces $\bar{y}_s$ with the single global mean ($\mu_{\text{global}}=0.645$) as a prior-pull control (if sub 19 ties sub 13, per-subject identity is irrelevant; if it loses, the per-subject signal is doing real work); sub 20 stacks sub 14’s shrunk-capped global K-shift onto sub 13. **Sub 17 scored -0.61 with AUC 0.69 (the highest of any submission), sub 25 scored $-0.61/0.68$, sub 18 scored -0.62 , and**

sub 19 scored -0.63 : the NLL minimum is a flat manifold spanning $\alpha \in [0, 0.5]$ (subs 17/21/13/25 all at -0.61) but the AUC has a strict concave peak at $\alpha=0.3$ (0.69, with symmetric 0.68 at $\alpha=0$ and $\alpha=0.4$, 0.67 at $\alpha=0.5$, 0.66 at $\alpha=0.7$). The content head, while contributing zero *NLL* on top of per-subject identity, does contribute *ranking* information — but only at a narrow low-weight band. Replacing per-subject identity with the global mean (sub 19) costs 0.02 NLL — the per-subject training-time mean $\bar{y}_s$ is the active ingredient. **Sub 20 scored** -0.63 : stacking the K-shift channel on top of the blend lost 0.02 NLL versus the pure blend but still beat sub 5 by 0.02, confirming once more that with only $K \times \text{categories} = 5 \times c$ labels the mean-residual estimator is too noisy to help even when the prediction distribution is non-flat.

Sub 21 ablation sweep (subs 23, 24, 26). To attribute sub 21’s gain we built three follow-ups: subs 23 ($\lambda=0.05$) and sub 24 ($\lambda=0.20$) bracket the smoothing parameter above and below sub 21’s $\lambda=0.10$, and sub 26 wraps sub 21 in a per-subject Beta-Bernoulli posterior update using the $K=5$ revealed labels as a likelihood (prior strength $n_s=25$, giving K an effective weight of $\approx 17\%$). **All three score identically to sub 21:** -0.61 NLL, AUC 0.68. The smoothing manifold is flat across a $4\times$ range of λ (matching the analogous flat *T*-ridge in Appendix A.1), and the adaptive $K=5$ channel contributes nothing detectable even when used in its theoretically right form (Bayesian posterior over Bernoulli observations, not additive residual shift). Combined with sub 8/14/20 (*K*-shift hurts or no-ops), this nails the F6 column shut: the leaderboard does not contain enough information in five revealed labels to update a 745-subject training-time accuracy lookup.

Strategies not pursued. F2 (IRT-only) fails cold-start by design: with no items in common with training, the difficulty parameters (a_j, b_j) have no support and reduce to a global prior ?. F5 (local LLM judge) does not fit the Codabench ? per-tier sandbox compute budget. Both are marked $\times$ in Table 2.

A.4 Error analysis: three compounding mechanisms

(i) Platt sharpening amplifies OOD errors. Our fitted Platt slope $\alpha \approx 1.15$ is greater than 1, meaning Platt *sharpens* the predicted probabilities slightly. This reduces in-distribution NLL but pushes any confidently-wrong prediction further toward the $\{0, 1\}$ boundary. We had already observed this on our own cold-start hold-out (raw $-0.83 \rightarrow$ Platt -0.88); the leaderboard appears to lie further OOD than our hold-out, so the same effect is presumably larger there. Sub 2 (identity Platt) recovered 0.08 NLL.

(ii) Capacity-induced overfitting. The encoder upgrade took us from 17M to 110M parameters and replaced sparse one-hots with 768-d benchmark and condition embeddings. With the same training data and the same MLP head shape, this gives the model many more degrees of freedom for fitting training-distribution-specific patterns. The 40976e8 submission’s lower capacity may have been acting as implicit regularization toward more transferable solutions.

(iii) Subject offset is noisy in 5-shot. The mean-residual offset shifts *all* of a subject’s predictions by a single constant computed from $K=5$ labels. If those five labels happen to land on items where the model is catastrophically wrong, the constant moves already-good predictions in the wrong direction. The 40976e8 submission’s Newton–Raphson update adjusted θ rather than the output, which was, in principle, more robust to such mismatches. Sub 3 (offset disabled) recovered another 0.08 NLL.

B Implementation Details

B.1 Full model architecture

The 2PL IRT backbone $P(R_{ij}=1 \mid \theta_i, a_j, b_j) = \sigma(a_j(\theta_i - b_j))$ is fit on the dense submatrix (≥ 5 observed responses per item; $\sim 33K$ items) by NaN-masked BCEWithLogitsLoss; 1000 epochs of Adam ($\eta = 10^{-2}$, wd = 0) on a T4 GPU. The shared encoder is sentence-transformers/all-mpnet-base-v2 (110M params, 768-d output), run in eval mode with mean-pooling. The 3075-d input $[\theta, \mathbf{e}_{\text{subj}}, \mathbf{e}_{\text{item}}, \mathbf{e}_{\text{bm}}, \mathbf{e}_{\text{cond}}, \mu_{\text{subj}}, \mu_{\text{bm}}]$ is consumed by the MLP in Table 5; training uses Adam ($\eta = 3 \times 10^{-3}$, wd = 10^{-4} , batch 2048), 30 epochs, NaN-masked

BCE loss, ReduceLROnPlateau on the in-distribution Platt hold-out, early stop on the cold-start hold-out.

Table 5: 5-layer MLP head used by sub 5.

Layer	Shape	Notes
Linear 1	3075 $\rightarrow$ 512	LayerNorm, ReLU, Dropout 0.2
Linear 2	512 $\rightarrow$ 256	LayerNorm, ReLU, Dropout 0.2
Linear 3	256 $\rightarrow$ 128	LayerNorm, ReLU, Dropout 0.1
Linear 4	128 $\rightarrow$ 64	LayerNorm, ReLU
Linear 5	64 $\rightarrow$ 1	raw logit

Calibration. Platt fits a 1-d logistic regression $\sigma(\alpha\ell + \beta)$ on the 10% in-distribution hold-out by L-BFGS. Temperature is fit on the cold-start hold-out by `scipy.optimize.minimize_scalar` over $T \in [0.5, 10]$. Final inference is $\hat{p} = \sigma((\alpha\ell + \beta)/T) - \delta_s$, clipped to $[10^{-3}, 1 - 10^{-3}]$.

B.2 Training pipeline, compute, and artifacts

Training runs on a personal Modal workspace (ap-2KtFZJ7o0hi57ZvqwvsMTc), single NVIDIA T4 GPU, 64 GB RAM. Wall-clock $\approx$ 20–25 min per run. Labels are binarized via $y_{ij} = \mathbb{I}[\text{score} \geq 0.5]$. The cold-start split holds `bfcl` and `matharena` out of *all* training (IRT, MLP, Platt); the Platt hold-out is a 10% uniform random hold-out of the remaining rows. Memory engineering: the dense submatrix is streamed shard-by-shard from parquet to avoid materializing the full $909 \times 104\text{K}$ matrix in RAM; embeddings are cached to disk per text field. Artifacts shipped in the submission ZIP: `mlp.pt`, `platt.json`, `temperature.json`, `theta_lookup.json`, `subject_emb_index.json`, `subject_embeddings.npy`, `benchmark_emb_index.json`, `benchmark_embeddings.npy`, `condition_emb_index.json`, `condition_embeddings.npy`, `subject_mean_acc.json`, `benchmark_mean_acc.json`, `global_mean_acc.json`, `centroids.npy`.

B.3 Dataset preprocessing

Table 6: Training-data composition and downstream splits.

Quantity	Count
Subjects	909
Unique items	103,983
Benchmarks (train / cold-start hold-out)	13 / 2
Cold-start held-out benchmarks	<code>bfcl</code> , <code>matharena</code>
Cold-start hold-out rows	397,484
Platt hold-out rows (10%)	303,543
MLP training rows	2,731,896

Subject text. Concatenation of the subject’s HuggingFace identifier, organization, and any free-text description, normalized by lowercasing and stripping `<eos>`/`<pad>` tokens.

Condition normalization. Conditions are stripped of trailing whitespace and collapsed to lower-case; the resulting 48 canonical strings are embedded by the shared encoder.

Filtering. Rows with NaN labels are dropped before binarization; subjects with < 5 total responses are kept but receive the global ability prior $\theta = 0$.

Feature standardization. μ_{subj} and μ_{bm} are computed on the MLP training rows only (i.e. excluding both the cold-start and Platt hold-outs) to prevent leakage; both are clipped to $[10^{-3}, 1 - 10^{-3}]$ before consumption.

C AI Disclosure

C.1 Plagiarism, Bias, and Inaccuracies Statement

We used only the public Hugging Face dataset `aims-foundations/measurement-db`, with no additional curated or scraped data. We did not redistribute its contents; all derived artifacts (embeddings, mean-accuracy tables, MLP weights) were computed in-house from the published parquet shards. We are aware that the response matrix is biased toward English-language items and toward subjects from a handful of large U.S./European labs, and we treat the resulting predictions accordingly: any inference about the performance of an underrepresented subject on an underrepresented benchmark is unreliable, and our cold-start hold-out (`bfc1 + matharena`) was chosen to expose, not hide, this limitation. We made no claims of correctness about specific items: a $\hat{p}$ value is a calibrated probability, not a verification of the underlying item–subject pair. Code and artifacts are available at the repository linked from the title page.

C.2 AI Tools Reflection

We used the GitHub Copilot CLI as a paired engineering and writing assistant throughout this project. Concretely, AI assistance helped us (a) extract and summarize the Discussion 6 tutorial slides so that we could plan the Part V ablation grid; (b) translate the textual calibration recipes into runnable Python variants for the `predict()` interface; (c) draft this report’s LaTeX and run syntactic checks. Each AI-generated edit was reviewed and either rewritten or accepted by a human author, and the underlying model artifacts (mpnet embeddings, MLP weights, IRT θ , temperature scalar) were trained and verified by us rather than produced by a language model. Going forward, we expect to keep using AI for the same role—fast pattern-matching against published technique catalogs—while reserving the experimental design, error analysis, and writing of insight to ourselves.

C.3 Impact Statement

Predictive evaluators like the one we built lower the cost of deciding which AI subject to deploy on which task, which is double-edged: they speed up beneficial deployments but can also encourage skipping direct measurement when a predicted probability “looks good enough.” We mitigate this in three ways. First, our final model deliberately publishes *flat* confidence ($T^* \approx 4$ effectively narrows every sigmoid output toward 0.5), so a user reading our predictions cannot mistake them for sharp claims about correctness. Second, the cold-start hold-out in §3.2 establishes the achievable NLL ceiling (≈ -0.66); any downstream user who needs sharper confidence should run the actual evaluation instead of trusting the predictor. Third, our code adheres to the competition’s network-isolation rule: no test-time data leaves the sandbox, no external APIs are queried, and no subject- or item-level data persists across rounds.